

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain :

- a) Cost economics and its effect on plastic product design
- b) Product life cycle and its stages

Q.24 Write short note on :

- a) Various materials and their selection criteria
- b) Explain gate side and its location

Q.25 Explain various types of welding methods and their types with the help of neat sketch.

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Roll No.

6th Sem/ Plastic Technology Subject : Plastic Product Design

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Alphabets and Letters are generally used for ____.

- a) Identification of shift
- b) For displaying company logos
- c) For texturing effect
- d) All of these

Q.2 The plastic should have ____ wall thickness.

- a) Uniform
- b) Non-Uniform
- c) Irregular
- d) None of these

Q.3 What should be the minimum distance between two consecutive holes ____.

- a) 4xd
- b) 2xd
- c) 3xd
- d) None of these

- Q.4 Minimum radius given in fillets for any plastic product design is _____.
 a) 3/8 inch b) 1/2 inch
 c) 1/4 inch d) 1 inch
- Q.5 Which part of the moulding process determines the diameter and wall-thickness of product?
 a) Molten Plastic b) Die Gap
 c) Mold Cavity d) Blow Pin
- Q.6 What must be avoided while designing mould component in blow moulding?
 a) Radii b) Bend
 c) Fillet d) Sharp corners

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Give two advantages of Ribs.
- Q.8 Name two permanent joining methods used in plastic assembly.
- Q.9 Dope cement generally contains _____ % of parent resins.

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- Q.10 Line formed by meeting of two parallel flow fronts is known as _____.
- Q.11 Gates should be located at the corners to provide longitudinal flow to material to reduce problem of shrinkage. (True / False)
- Q.12 Name different type of undercuts.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain positioning of holes.
- Q.14 State various stages of new product development.
- Q.15 Explain importance of wall thickness in plastic design.
- Q.16 Write short note on shapes used in plastic product designing.
- Q.17 Explain taper and draft angle.
- Q.18 Define texturing and various techniques for its manufacturing.
- Q.19 Write short note on undercuts.
- Q.20 Discuss tolerance and its types.
- Q.21 Explain role of ribs and bosses in plastic product design.
- Q.22 Discuss moulded inserts.

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